COVER FEATURE

India's semiconductor moment has been "imminent" for so long that it risks sounding like a tech journalist's inside joke. And yet, 2026 feels palpably different. Not because the country has suddenly cracked leading-edge fabs or outspent the incumbents, but there's visible momentum and signs of things gathering pace despite all the insurmountable challenges related to building a semiconductor manufacturing ecosystem – and everything it encapsulates.

Across the following pages, you'll hear from the architects of this ecosystem. Together, their perspectives reveal a more mature narrative. India's semiconductor journey will not be defined by a single mega-fab ribbon-cutting moment. It will be defined by design talent density, materials science, EDA depth, packaging innovation, patient capital. And most importantly, a willingness to collaborate globally while building locally.

If the last decade was about India consuming silicon at scale, the next will test whether we can create it with intent. This cover story is less a celebration than a reality check. And perhaps, finally, the outline of a concrete roadmap.

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INDIA'S SEMICONDUCTOR STORY

1976–1984

India's first fab dream begins
The Indian government approved Semiconductor Complex Limited (now SCL) in 1976, with production starting in 1984 in Mohali. It was one of Asia's earliest chip fabrication attempts before even TSMC, producing CMOS chips through a technology collaboration with American Microsystems.

1989–1997

Fire, delays and lead lost
A major 1989 fire destroyed much of the SCL fab, stalling India's early manufacturing momentum. Production restarted in 1997 but at a limited scale, while Taiwan and South Korea surged ahead commercially.

MID-1980s–2000s

World's chip design back-office
Texas Instruments set up India's first semiconductor design centre in Bengaluru in 1985, triggering a wave of global R&D centres. Over the next three decades, India evolved into a major global hub for VLSI design and chip engineering talent.

2000s–2010s

Research and talent pipelines
Government-backed nanoelectronics programmes, Centres of Excellence and academic-industry collaborations built strong semiconductor R&D capability and design expertise across India.

2021

India Semiconductor Mission begins
In December 2021, India launched the ₹76,000-crore India Semiconductor Mission to support fabs, display manufacturing, packaging and chip design, aiming to build a full domestic semiconductor ecosystem.

2023–2025

'Made-in-India' chips and ecosystem buildout
India began unveiling indigenous chips and approving multiple fab, OSAT and design projects under the mission. The country's first domestically developed chips and pilot production milestones were showcased at Semicon India events.

2025–2026

Commercial production era begins
Pilot production has already started in several semiconductor plants, with commercial chip production expected to begin from 2026 as multiple fabs and packaging units come online.